

Hyeonsu Jeong

Undergraduate Researcher | Department of Industrial and Systems Engineering

Gyeongsang National University

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EDUCATION

03/2021 – 02/2027 **B.E. in Industrial and Systems Engineering**
 B.E. in Big Data Convergence (Double)
 GPA: 3.43/4.5
 Gyeongsang National University, Jinju, South Korea

09/2024 – 07/2025 **Smart Manufacturing Big Data Micro-Degree**
 Gyeongsang National University, Jinju, South Korea

RESEARCH EXPERIENCE

06/2025 – 02/2026 **Undergraduate Researcher**
 Digital Transformation Laboratory, Gyeongsang National University, Jinju, Korea
 - Developed optimization models for visual sensor placement (Greedy, SA, LAHC)

RESEARCH PROJECTS

▪ Lead Researcher

10/2025 – 01/2026 **Optimization of Multi-Type CCTV Camera Placement for Manufacturing Site Monitoring**
 Gyeongsang National University – National University Development Program

06/2025 – 12/2025 **AI-based CCTV Image Analysis for Worker Safety Management on Manufacturing Sites**
 Gyeongsang National University – RISE Industry-Academia Collaborative Project

▪ Research Assistant

06/2025 – 12/2025 **Digital Twin Platform for Smart Construction and Automation**
 National Research Foundation of Korea (NRF) – STEAM Research Project (Strategic Program)

PUBLICATIONS & RESEARCH OUTPUTS

▪ **Journal Articles**

- [1] **Jeong, H.**, Lee, S., and Moon, S.*. “Automated Sensor Location Optimization for Efficient Visual Data Acquisition for Digital Twin-powered Construction Monitoring.” *Advanced Engineering Informatics*. [Manuscript under editorial discussion, Q1, SCI-E, 2025 IF 9.9]

*Corresponding Author

▪ **Conference Proceedings**

- [1] **Jeong, H.**, Lee, S., and Moon, S.* (2025). “Optimization of CCTV Camera Placement for Construction Site Monitoring.” *Oral Presentation, 2025 Fall Conference of Korea Institute of Industrial Engineers (KIIE)*, Daejeon, South Korea.
- [2] **Jeong, H.**, Lee, S., and Moon, S.* (2025). “Development of Foundational Technology for AI-based CCTV Video Analysis for Worker Safety Management on Manufacturing Sites.” *Poster Presentation, 2025 The Next AI (Gyeongnam)*, Changwon, South Korea.

*Corresponding Author

▪ **Intellectual Property Rights**

Patent (Registered)

- [1] **Jeong, H.**, Moon, S., and Kim, J. (2026. 04. 23.) “Method for Exploration on Optimal Locations of CCTV Cameras for Construction Site Monitoring.” Application No. 10-2026-0073586. (Korea)

Software

- [1] Lee, S., Moon, S., **Jeong, H.**, and Kim, D. (2025. 9. 11.) “Program for Realistic Color Matching on 3D Point Clouds of Industrial Sites.” (Korea, Program C-2025-035569)

RESEARCH INTERESTS

- Optimization and Decision-Making for Complex Systems
- Spatial Optimization and Resource Allocation in Real-world Environments
- Computer Vision-based System Modeling and Data-driven Optimization
- Intelligent Monitoring Systems and Sensor Placement Optimization

AWARDS

- | | |
|---------|--|
| 04/2026 | 2026 GSAT Young Startup Camp Competition (Excellence Prize)
Awarded by CEO, Gyeongnam Creative Economy Innovation Center |
| 02/2026 | GNU RISE Shared Achievement Forum – Outstanding Startup Club (Commendation)
Awarded by Director, GNU RISE Program |

12/2025	Southeast Regional University Startup Competition (Grand Prize) Awarded by Director of Startup Support Center, Gyeongsang National University
11/2025	Data Venture Problem-Solving Challenge (Excellence Award) Awarded by Director of Big Data Innovation Convergence University, Seoul National University
11/2025	GNU Pioneer Leaders Startup Idea Hackathon (Grand Prize) Awarded by Director, GNU Startup Support Center
09/2025	Gyeongnam Regional University Startup Club Competition (Grand Prize) Awarded by Director, GNU Startup Support Center
07/2025	GNU-Startup Pioneer (Excellence Award) Awarded by Director, GNU Startup Education Innovation Center
06/2025	Convergence Data Utilization Startup Competition (Public Institution Head Award) Awarded by President of Korea SMEs and Startups Agency
05/2025	GNU Pioneer Leaders Startup Ideathon (Grand Prize) Awarded by Director, GNU Startup Support Center
01/2025	ICT Convergence Competition (Generative AI Track) (Grand Prize) Awarded by President of Gyeongsang National University
01/2025	Winter Big Data Tableau Competition (Encouragement Prize) Awarded by CEO, Planit Partners
12/2024	AWS GenAI Schumpeter Hackathon (Grand Prize) Awarded by CEO of AWS Korea
12/2024	AWS GenAI Schumpeter Rocket Pitch (Excellence Award) Awarded by CEO of AWS Korea
12/2024	Creative Convergence 3D Printing Idea Competition (Encouragement Prize) Awarded by Dean, GNU College of Engineering
09/2024	National University Development Program Student Project Competition (Excellence Award) Awarded by Director, GNU Dept. of Industrial & Systems Engineering

ACTIVITIES

▪ Experience

03/2026 – Present	Startup-Centered University Program: “Worker-Centric Industrial Safety Monitoring and Legal Decision Support Platform.” (Grant Number: 20410310) Korean Government-Funded Startup Program
12/2025 – 01/2026	Experiential Project-Based Learning Program for International Students: Face Recognition Attendance System FPT IS & FPT University, Danang, Vietnam.
12/2025	2025 Gyeongnam Global Innovation Festa G-NEX – Startup Booth Operation Gyeongnam Global Innovation Festa
10/2025	Student Startup Team U-300+ – National Finals Qualifier Ministry of Education, Korea

- 01/2025 **Lecture at Next-Generation Startup School**
Main Contents Co., Ltd.
- 11/2024 – 02/2025 **GNU Pioneer Leaders Startup Supporters, 1st Cohort**
GNU Startup Support Center, Gyeongsang National University

▪ **Leadership**

- 05/2025 - ~ **CEO, Startup “Workrithm”**
Jinju, Korea.
- 01/2026 – 12/2027 **President, Startup Club “APLUSES”**
Gyeongsang National University, Jinju, Korea.
- 03/2025 – 02/2026 **Planning Director, Startup Club “APLUSES”**
Gyeongsang National University, Jinju, Korea

TECHNICAL SKILLS

▪ **Optimization Algorithms**

- Greedy Algorithm, Simulated Annealing (SA), Late Acceptance Hill Climbing (LAHC)

▪ **Programming & Backend**

- Python, FastAPI
- Kotlin (Android Studio)

▪ **Data & Systems**

- Cloud Compare (Point Cloud Data Processing)
- Neo4j (Graph Database), Graph Modeling
- Data preprocessing and pipeline construction

▪ **AI / Computer Vision**

- CCTV-based monitoring system development
- Face recognition (ArcFace, RetinaFace)

PROFESSIONAL CERTIFICATES AND TRAINING

- 2025 **SQL Developer (SQLD)**
Korea Data Agency (KDATA), Seoul, South Korea
- 2025 **Drone License (Level 4)**
Korea Data Certification Authority
- 2024 **Certificate of Computer Literacy (Level 2)**
Human Resources Development Service of Korea